Not All Goals Are Equal: A Time-Aware Expected-Points Framework for Goal Valuation

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Abstract

The standard scoreboard treats every goal as one unit of contribution, indifferent to the gamestate it altered or the minute it landed. A consolation goal at 4–0 down is filed under the same column as a 90th-minute equaliser. This project builds a time-aware expected-points (EP) framework that prices every goal by both dimensions: the `(gd_before, minute_bin)` pre-goal state it transitioned out of and the `(gd_after, minute_bin)` post-goal state it transitioned into. Across **39,381 goals** in six top-flight European leagues over seven seasons, the time-aware model produces a substantively different player ranking from the time-blind baseline. The most valuable single transition observed is the **0→1 in the 75–90 minute bin (1.68 expected points collected, 2.14× the same transition in 0–15 minutes)**. Mário Rui (Napoli, 28 G+A) is the model's biggest gainer at **+3.43 ΔEP** between the timed and untimed rankings — a left-back whose involvements, mostly assist-type credits rather than goals, land unusually often in high-value minutes the time-blind model underrates.

1. Research Question

> *Does conditioning a goal's expected-points value on the minute it was scored — in addition to the goal-difference transition it produced — yield a materially different ranking of players from the time-blind baseline, and does that difference cleanly identify players whose contributions cluster in high-leverage minutes?*

This is not a hypothesis test. The framework rests on two evaluable claims:

1. **Estimation:** the empirical state-value table `EP(gd, minute_bin)` is well-conditioned given the sample size in each cell.
2. **Interpretability:** the timed-vs-untimed delta on the player leaderboards reshuffles in directions a domain expert would predict (late equalisers ↑, garbage-time goals ↓).

Both are addressed in §4.

2. Data

Sources: Per-match shot-event tables, FBref-derived, consolidated into one pickle per league. Files named `match_data_final_<league>.pkl` in `RENBALL_MATCH_DATA_PATH`. Files containing `CHECK` in the basename are excluded.

Leagues (6): Premier League, Serie A, Bundesliga, La Liga, Ligue 1, Eredivisie.

Seasons (7): 2017-2018 through 2023-2024.

Sample: 39,381 goal events across the 6 × 7 league-season cells.

Per-event variables used:

FIELD	SOURCE COLUMN (AFTER `_NORMALIZE_COLS`)	NOTES
Scoring team	`…__Squad`	Disambiguated against the inferred home/away pair
Outcome	`…__Outcome`	Filtered to `"goal"`
Goalscorer	`…__Player`	Cleaned by `_clean_player_name` (strips `"(pen)"` etc.)
SCA1 (assist)	`SCA 1__Player`	Same cleaning
Minute (raw)	`…__Minute`	May include stoppage suffix `45+1`, `90+3`

Cleaning decisions:

1. Drop matches with `error == "Failed to retrieve data after repeated attempts."`
2. Drop matches where the shot table (`shots.shots_all`) is missing or empty
3. Drop goal events where the scoring team cannot be matched to the inferred home/away team pair (rare; ~0.1% of events)
4. **Parse minute strings:** `"45+1"` → 46, `"90+3"` → 93. **Stoppage-time rules:** any `"45+X"` falls in the `30-45` bin; any `"90+X"` (or `minute > 90`) falls in the `75-90` bin
5. Goal events with unparseable minutes are assigned to an `unknown` bin and excluded from all timed analyses
6. The editorial leaderboards drop `"Unknown"` contributors — almost all of them goals with no recorded SCA1 (unassisted goals; own goals are not in the shot table at all). They aggregate to a fake "player" who would otherwise top the time-aware leaderboard with ~2,250 events. The Tabulator dashboard retains them for transparency.

3. Methodology

3.1 State space

- **Goal differences (15 states):** integer in `[-7, +7]` from the scoring team's perspective
- **Minute bins (6 states):** `0-15`, `15-30`, `30-45`, `45-60`, `60-75`, `75-90`
- **Pre-goal state space size:** 15 × 6 = **90 theoretical states; 82 observed empirically** (8 extreme `(gd, bin)` combinations had no events)
- **One-step transitions kept:** `{(gd, gd+1) for gd in [-7, +7]}` (15 transitions, scoring team's perspective)

3.2 Time-bin rationale

A continuous-time model (minute-by-minute) would over-fit at the seam minutes 45 and 90, where stoppage muddies the assignment. 15-minute bins are **coarse enough** to keep each cell well-populated (median bin n ≫ 1,000 in the central GD rows) and **fine enough** to detect the late-game value premium — the headline result is that the 0–15 vs 75–90 difference for 0→1 is 2.14×, which a 2-bin (early/late) cut would still surface but more granular phases (e.g. how 30–45 differs from 45–60) would not.

3.3 State value function (`EP(state)`)

For each pre-goal state `(gd, minute_bin)`:

1. **Symmetric expansion of events.** Each goal event in the dataset produces two pre-goal-state rows: the scoring team gets a row with `(gd_before, minute_bin)`, the conceding team gets a row with `(-gd_before, minute_bin)`. Each row carries the **owning team's** final result — scorer rows use the scoring team's `final_points`, conceder rows use the conceder's (W↔L / 3↔0 flip applied; draws stay D / 1 point). This is critical: without the flip, both rows would carry the scoring team's outcome and the table would alias the two sides together, so `EP(gd=+7)` and `EP(gd=-7)` would come out equal instead of mirror images.
2. **Empirical distribution of final results.** Group by `(gd, minute_bin)`, compute `pW`, `pD`, `pL` as the value-counts normalised across that cell's rows.
3. **Expected final points:** `EP(gd, minute_bin) = 3·pW + 1·pD`. Implementation uses the mean of `final_points` over the group, which is mathematically identical and numerically more stable.

The time-agnostic state value `EP_untimed(gd)` is computed identically except the grouping drops `minute_bin`.

The state-value table on the editorial page shows `EP(gd, minute_bin)` for `gd ∈ [-4, +4]` (the populated rows). Numerically the table runs from ≈ 0 along the bottom-most row (team trailing badly with little time left) up to ≈ 3 along the top-most (team leading comfortably late), with the level-score row `gd = 0` settling near the long-run home/away mean of ~1.5 expected points.

3.4 Goal value (EP credit per event)

A goal is priced from the transitions themselves, not from the state table. For each one-step transition `g → g+1` in minute bin `b` (scoring team's perspective), let `EP_T(g → g+1, b)` be the mean final points of the scoring teams over every goal that made that transition in that bin.

- **Timed credit:** `EP_T(g → g+1, b) - EP_T(g-1 → g, b)` — the transition minus the previous observed transition in the same bin (in the populated rows, the one directly below it). It is the gap in final points between teams that reached `g+1` with a goal in that window and teams that reached `g`. The lowest observed transition in each bin has nothing to compare against and is set to 0.
- **Untimed credit:** the same difference with the minute bins pooled.

Implementation: `analysis.py`, `prep_transitions_timed` / `prep_transitions` — transitions sorted by goal difference within each bin, then `exp_points.diff()`, with the first row of each bin filled with 0.

The conceding team gets the negation of the same number. Only one-step transitions in the `[-7, +7]` GD range are retained; multi-step transitions (impossible in football without intervening events) and out-of-range states are dropped.

The full value map `val_map_timed[(gd_before, gd_after, b)] → ep_collected` and its untimed counterpart `val_map_untimed[(gd_before, gd_after)] → ep_collected` are the canonical lookup tables used downstream.

3.5 Attribution

Goals are credited to three different roles, producing three parallel contributor tables:

- **`Goalscorer`** — credit to the player who scored
- **`Assist`** — credit to the FBref SCA1 (first shot-creating action) for that shot
- **`Goals+Assists`** — additive pooling of both, the editorial-page default

Each table is built per `(scoring_team, contributor)` for every league-season, then rolled up into four views:

VIEW	AGGREGATION
`Per League & Season`	Raw per-season tables (42 sub-tables = 6 leagues × 7 seasons)
`Totals by League`	Sum across seasons within a league
`Totals by Season`	Sum across leagues within a season
`All`	Sum across both axes

The editorial-page leaderboards pool further to one row per `contributor`, picking the team where the player has the most goals_involved as the `primary_team` label. The Tabulator dashboard exposes the per-`(scoring_team, contributor)` rows for users who want to inspect transfers and joint affiliations.

Each contributor row also stores transition-resolved breakdowns: `cnt_<gd_before>_to_<gd_after>` for every kept transition, `cnt_bin_X` per time bin, and `ep_bin_X` per time bin (the slice of `exp_points_collected_timed` that came from goals in bin X).

3.6 Per-90 normalisation

Total EP rewards volume of involvement; **EP per 90 minutes** rewards rate. Total minutes played per player are pooled across all leagues and seasons in scope from the per-match player tables (FBref `ROWS_DICTS_SUMMARY` inside `FULL_DICT_ROWS.pkl`, top-5 leagues only — Eredivisie is not in this source and Eredivisie-only players are excluded from the per-90 leaderboard). For each pooled contributor row the build computes:

```
ep_per_90 = (exp_points_collected_timed / minutes_played) × 90
```

The editorial-page leaderboard applies a `minutes_played ≥ 900` filter (≈ ten full matches) to keep the ranking meaningful. The Tabulator dashboard surfaces `minutes_played` and `ep_per_90` as additional columns on every Goalscorer / Assist / Goals+Assists view; sorting by `ep_per_90` with the smart filter (e.g. `>0.4`) lets the reader apply their own threshold.

3.7 Pooled vs per-league treatment

The editorial page (`analysis.html`) uses pooled value functions — a single `EP(gd, minute_bin)` table built from all six leagues combined. The Tabulator dashboard (`dashboard.html`) exposes both pooled and per-league state-value and transition tables. Per-league tables share most of the qualitative features of the pooled table (the late equaliser ranks high in every league) but vary in magnitude at the extremes — a function of sample size more than league effect. The pooled choice for the editorial page is deliberate: the headline narrative is about the universal shape of the value function, not league-specific calibration.

3.8 Software stack

- Python 3.12 (project venv at repo root)
- `pandas` — data wrangling, group-by aggregation
- `numpy` — numeric primitives
- `openpyxl` — Excel sheet writing (local-only, not committed)
- Output: `data/data.json` (editorial payload) and `data/dashboard/` (18 per-`(dataset, view)` JSONs + manifest, sparse-encoded, each ≤ 4 MB)

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4. Results

> *All numbers in this section come from* `code/build.py` → `data/data.json` *and are surfaced live in the editorial page.*

4.1 Headline numbers

METRIC	VALUE
Total goal events	39,381
Modelled `(gd, minute_bin)` states	82 of 90
Best transition (most EP collected per event)	**0→1 @ 75–90: 1.68 EP**
Lowest-valued transition	-2→-1 @ 0–15: 0.00 EP (n=17) — the lowest observed transition in its bin, set to 0 by construction (§3.4)
Early-vs-late opener ratio	**2.14×** (1.68 / 0.79)

4.2 0⟶1 by minute bin (the cleanest single-transition story)

BIN	EP COLLECTED	N
0–15	0.79	4,359
15–30	0.74	3,539
30–45	0.84	2,998
45–60	1.05	2,367
60–75	1.22	1,934
75–90	**1.68**	2,300

4.3 Top 10 EP collectors (Goals + Assists, pooled)

#	PLAYER	PRIMARY TEAM	G+A	EP COLLECTED (TIMED)
1	Lionel Messi	Barcelona	267	157.75
2	Mohamed Salah	Liverpool	251	154.78
3	Kylian Mbappé	Paris S-G	274	154.53
4	Harry Kane	Tottenham	240	150.72
5	Robert Lewandowski	Bayern Munich	270	149.10
6	Ciro Immobile	Lazio	208	128.34
7	Iago Aspas	Celta Vigo	179	120.56
8	Cristiano Ronaldo	Juventus	174	119.25
9	Wissam Ben Yedder	Monaco	182	116.41
10	Karim Benzema	Real Madrid	174	110.76

4.4 Top 10 timed upgrades (largest positive ΔEP, min 5 G+A)

#	PLAYER	PRIMARY TEAM	G+A	ΔEP (TIMED - UNTIMED)
1	**Mário Rui**	**Napoli**	28	**+3.43**
2	Evann Guessand	Nice	19	+3.34
3	Cristian Tello	Betis	36	+2.82
4	Adama Traoré	Wolves	43	+2.69
5	Youssoufa Moukoko	Dortmund	22	+2.58
6	Viktor Kovalenko	Spezia	7	+2.53
7	Sasa Kalajdzic	Stuttgart	37	+2.50
8	Mario Hermoso	Atlético Madrid	20	+2.33
9	Leo Dubois	Lyon	26	+2.27
10	Lee Kang-in	Mallorca	31	+2.20

The upgrade leaderboard is led by defenders and attackers whose goal involvements fall disproportionately in high-value minutes. The list carries no uncertainty intervals, so small gaps between neighbours should not be read as a ranking.

5. Limitations

1. **No expected-goal weighting.** Every goal is treated as 1.0 — a 2-yard tap-in and a 30-yard volley contribute identically. An xG-weighted version would price by chance creation rather than chance conversion.
2. **Penalties are not separated, and own goals are missing.** Penalty goals trigger the same transition value as any other. Own goals do not appear in the shot tables at all, so final scores rebuilt from them are too low in some matches, and every goal after a missing own goal carries the wrong game state.
3. **Assist credit caps at SCA1.** FBref's SCA2 is recorded but unused. Build-up sequences distribute credit further upstream than the framework captures.
4. **Observational, not causal.** The state-value function is computed empirically against final results. It is a value table, not a causal model — it does not isolate the goal's effect from concurrent factors (red cards, substitutions, fatigue) that also influence the final outcome. The "value of a goal" is "the EP gap between the two states the goal connects" — readers should not interpret it as "the EP this goal added," which would require a counterfactual the data does not provide.
5. **Pooled value function across leagues and seasons.** The editorial-page rankings use the pooled values. Differences between, say, Eredivisie defensive trends and Premier League ones would mean a "0→1 at minute 80" goal carries slightly different empirical EP in each league. The dashboard surfaces league-specific tables for users who want to inspect this.
6. **Stoppage-time goals fold into the 75–90 bin.** A 95th-minute goal and a 78th-minute goal contribute to the same cell. A finer split would separate them but at the cost of sample size at the extremes.
7. **Player attribution does not track transfers within a season.** A player who scored 10 goals at one club and 5 at another in the same season appears as one row per `(team, player)` in the raw tables. The editorial leaderboards pool further to one row per player by picking the primary team; this loses the transfer signal that the dashboard retains.
8. **The model rewards proximity to the threshold, not the goal itself.** A late equaliser is worth nearly a full point because the team it credits was about to win zero. A goal valuation framework grounded in counterfactual reasoning (e.g. permutation tests against alternative goal-minute distributions for the same player) would distinguish the player's contribution from the situation's leverage. Future work.
9. **The state table is sampled at goal moments.** Each of its rows comes from a goal event, so `EP(gd, b)` describes teams in that state at the moment a goal is scored — by either side — not teams in that state at a random minute. The goal values (§3.4) are computed from the transitions, not from this table.

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6. References

- **Source data:** FBref (fbref.com). Match shot tables consolidated locally into `match_data_final_<league>.pkl`.
- **Interactive dashboard:** `dashboard.html` in this project folder
- **Editorial summary:** `analysis.html` in this project folder
- **Project page:** renball.com/projects/02-points-by-gamestate/

- **Source code:** `code/` directory — `config.py`, `extraction.py`, `analysis.py`, `dashboard.py`, `main.py`, `build.py`